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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- If two events A and B such that $P(A') = 0.3$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$, then $P\left(\frac{A}{A \cup B'}\right)$ is
(a) $\frac{1}{4}$ (b) $\frac{3}{8}$ (c) $\frac{1}{8}$ (d) None
- If $y = mx$ bisects the angle between the lines $x^2(\tan^2 \theta + \cos^2 \theta) + 2xy \tan \theta - y^2 \sin^2 \theta = 0$ when $\theta = \frac{\pi}{3}$, then the value of $\sqrt{3}m^2 + 4m$ is
(a) 1 (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) $7\sqrt{3}$
- If $f: R \rightarrow R$ and $g: R \rightarrow R$ and continuous functions, then the value of the integral $\int_{-\pi/2}^{\pi/2} [f(x) + f(-x)][g(x) - g(-x)] dx$ is
(a) π (b) 1 (c) -1 (d) 0
- The maximum value of $(\cos \alpha_1)(\cos \alpha_2) \dots (\cos \alpha_n)$ where $0 \leq \alpha_1, \alpha_2, \alpha_n \leq \frac{\pi}{2}$ and $(\cot \alpha_1)(\cot \alpha_2) \dots (\cot \alpha_n) = 1$ is
(a) $\frac{1}{2^{n/2}}$ (b) $\frac{1}{2^n}$ (c) $\frac{1}{2n}$ (d) 1
- Let M be a point inside the ΔABC . Then, which one of the following is true?
(a) $AB + AC < MB + MC$ (b) $AB + AC > MB + MC$
(c) $AB + AC = MB + MC$ (d) None of these
- A line L has intercepts 'a' and 'b' on the coordinate axes. When the axes are rotated through a given angle, keeping the origin fixed, the same line has intercepts 'p' and 'q' which of the following statements is true?
(a) $a^2 + b^2 = p^2 + q^2$ (b) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$
(c) $a^2 + p^2 = b^2 + q^2$ (d) $\frac{1}{a^2} + \frac{1}{p^2} = \frac{1}{b^2} + \frac{1}{q^2}$
- If a, b are the roots $x^2 + px + 1 = 0$ and c, d are roots of $x^2 + qx + 1 = 0$ the value of $E = (a - c)(b - c)(a + d)(b + d)$ is
(a) $p^2 - q^2$ (b) $q^2 - p^2$
(c) $q^2 + p^2$ (d) None of these
- If $f(x) + f(1 - x) = 2$, then the value of $f\left(\frac{1}{2001}\right) + f\left(\frac{2}{2001}\right) + \dots + f\left(\frac{2000}{2001}\right)$ is
(a) 2000 (b) 2001 (c) 1999 (d) 1998
- Suppose a, b, c are in AP with common difference d. then, $e^{1/c}, e^{b/ac}, e^{1/a}$ are in
(a) AP (b) GP (c) HP (d) None
- Let α and β be the roots of the equation $x^2 + x + 1 = 0$. The equation whose roots are α^{19} and β^7 is
(a) $x^2 - x - 1 = 0$ (b) $x^2 + x - 1 = 0$
(c) $x^2 - x + 1 = 0$ (d) $x^2 + x + 1 = 0$
- In the expression $(x + 1)(x + 4)(x + 9)(x + 16) \dots (x + 400)$ the coefficient of x^{19} is
(a) 2870 (b) 210 (c) 4001 (d) 1900
- The value of $y = 0.36 \log_{0.25} \left(\frac{1}{3} + \frac{1}{3^2} + \dots\right)$ is
(a) 0.9 (b) 0.18 (c) 0.6 (d) 0.25
- If $H_1, H_2, \dots, H_n$ are n harmonic means between a and b, $a \neq b$, then the value of $\frac{H_1 + a}{H_1 - a} + \frac{H_n + b}{H_n - b}$ is equal to
(a) $n + 1$ (b) $n - 1$ (c) $2n$ (d) $2n + 3$
- For $a > 0, a \neq 1$, the number of values of x satisfying the equation $2 \log_x(a) + \log_{ax}(a) + 3 \log_{a^2x}(a) = 0$
(a) 2 (b) 3 (c) 4 (d) 5
- An eight digit number divisible by 9 is to be formed by using 8 digits out of the digits 0, 1, ..., 9 without replacement. The number of ways in which this can be done is
(a) 9! (b) $2(7!)$ (c) $4(7!)$ (d) $36(7!)$
- The number of ordered pairs $(m, n), m, n \in \{1, 2, \dots, 100\}$ such that $7^m + 7^n$ is divisible by 5 is
(a) 1250 (b) 2000 (c) 2500 (d) 5000
- If a, b, c are the roots of the equation $x^2 - 3px^2 + 3qx - 1 = 0$, then the centroid of the triangle with vertices $\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right)$ and $\left(c, \frac{1}{c}\right)$ is at the point
(a) (p, q) (b) $\left(\frac{p}{3}, \frac{q}{3}\right)$
(c) $(p + q, p - q)$ (d) $(3p, 3q)$
- Equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x-axis is
(a) $\sqrt{3}y = 3x + 1$ (b) $\sqrt{3}y = -(x + 3)$
(c) $\sqrt{3}y = x + 3$ (d) $\sqrt{3}y = -(3x + 1)$
- The number of roots of the equation $|x^2 - x - 6| = x + 2$ is
(a) 2 (b) 3 (c) 4 (d) None
- A pair of unbiased dice is rolled together till a sum of either 5 or 7 is obtained. The probability that 5 comes before 7 is
(a) $3/5$ (b) $2/5$ (c) $4/5$ (d) None
- A letter is taken at random from the letters of the word 'STATISTICS' and another letter is taken at random from the letters of the word 'ASSISTANT'. The probability that they are the same letters is
(a) $\frac{1}{45}$ (b) $\frac{13}{90}$ (c) $\frac{19}{90}$ (d) $\frac{5}{8}$
- A bag contains 6 red and 4 green balls. A fair dice is rolled and a number of balls equal to that appearing on the dice is chosen from the urn at random. The probability that all the balls selected are red is
(a) $\frac{1}{3}$ (b) $\frac{3}{10}$ (c) $\frac{1}{8}$ (d) None
- The value of λ for which the volume of parallelopiped formed by the vectors $i + \lambda j + k, j + \lambda k$ and $\lambda i + k$ is minimum is given by
(a) -3 (b) 3 (c) $\frac{1}{\sqrt{3}}$ (d) $-\sqrt{3}$
- A six faced die is a biased one. It is thrice more likely to show an odd number than to show an even number. It is thrown twice. The probability that the sum of the numbers in the two throws is even is
(a) $4/8$ (b) $5/8$ (c) $6/8$ (d) $7/8$



25. A letter is known to have come from either TATANAGAR or CALCUTTA. On the envelope, just two consecutive letters, TA, are visible. The probability that the letter has come from CALCUTTA is
 (a) $\frac{4}{11}$ (b) $\frac{1}{3}$ (c) $\frac{5}{12}$ (d) None
26. If $\cos \alpha + \cos \beta = a$, and $\sin \alpha + \sin \beta = b$ and θ is the arithmetic mean between α and β , then $\sin 2\theta + \cos 2\theta$ is equal to
 (a) $\frac{(a+b)^2}{(a^2+b^2)}$ (b) $\frac{(a-b)^2}{(a^2+b^2)}$
 (c) $\frac{a^2-b^2}{a^2+b^2}$ (d) None of these
27. If $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$, then the value of n is
 (a) 21 (b) 22 (c) 23 (d) 24
28. The value of $\sin 12^\circ \sin 48^\circ \sin 54^\circ$
 (a) $\sin 30^\circ$ (b) $\sin^2 30^\circ$
 (c) $\sin^3 30^\circ$ (d) $\cos^3 30^\circ$
29. The value of λ such that the four points whose position vectors are $3\hat{i} - 2\hat{j} + \lambda\hat{k}$, $6\hat{i} + 3\hat{j} + k$, $5\hat{i} + 7\hat{j} + 3\hat{k}$ and $2\hat{i} + 2\hat{j} + 6\hat{k}$ are coplanar is
 (a) -6 (b) 4 (c) 5 (d) 8
30. Let $A = 2\hat{i} + \hat{j} - 2\hat{k}$ and $B = \hat{i} + \hat{j}$. If C is a vector such that $A \cdot C = |C|$, $|C - A| = 2\sqrt{2}$ and the angle between $A \times B$ and C is 30° , then $|(A \times B) \times C|$ is equal to
 (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) 2 (d) 3
31. A rigid body is rotating at the rate of 3 rads about an axis AB, where A and B are the points $(1, -2, 1)$ and $(3, -4, 2)$. The velocity of the point P at $(5, -1, -1)$ of the body is
 (a) $3\hat{i} + 8\hat{j} + 10\hat{k}$ (b) $\frac{3\hat{i} + 8\hat{j} + 10\hat{k}}{3}$
 (c) $\frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}$ (d) $4\hat{i} + \hat{j} + 2\hat{k}$
32. If $f(x)$ is a spolynomial satisfying $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and $f(3) = 28$, then $f(4)$ is given by
 (a) 63 (b) 65 (c) 67 (d) 68
33. If $A + B + C = 0$, $|A| = 3$, $|B| = 5$, $|C| = 7$, then the angle between A and B is
 (a) $\frac{\pi}{6}$ (b) $\frac{2\pi}{3}$ (c) $\frac{5\pi}{3}$ (d) $\frac{\pi}{4}$
34. Suppose $P_1, P_2, \dots, P_{30}$ are thirty sets each having 5 elements and $Q_1, Q_2, \dots, Q_n$ are n sets with 3 elements each. Let $\bigcup_{i=1}^{30} P_i = \bigcup_{j=1}^n Q_j = S$ and each element of S belongs to exactly 10 of the P_i s and exactly 9 of the Q_j s. Then n is equal to
 (a) 15 (b) 3 (c) 45 (d) None
35. The number of functions f from the set $A = \{0, 1, 2\}$ into the set $B = \{0, 1, 2, 3, 4, 5, 6, 7\}$ such that $f(i) \leq f(j)$ for $i < j$ and $i, j \in A$ is
 (a) 8C_3 (b) ${}^8C_3 + 2({}^8C_2)$
 (c) ${}^{10}C_3$ (d) None of these
36. The value of $\int_0^{\pi/2} \frac{dx}{1 + \tan^3 x}$ is
 (a) 0 (b) 1 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

37. The integer n for which $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number is
 (a) 1 (b) 2 (c) 3 (d) 4
38. The area of the plane bounded by the curves $y = \sqrt{x}$, $x \in [0, 1]$, $y = x^2$, $x \in [1, 2]$ and $y = -x^2 + 2x + 4$, $x \in [0, 2]$ is
 (a) $\frac{10}{7}$ sq. units (b) $\frac{19}{3}$ sq. units
 (c) $\frac{3}{5}$ sq. units (d) $\frac{4}{3}$ sq. units
39. The function $f(x) = 2 \sin x + \sin 2x$, $x \in [0, 2\pi]$ has absolute maximum and minimum at
 (a) $\frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\frac{\pi}{3}, \pi$ (c) $\frac{5\pi}{3}, \pi$ (d) None
40. If $y = \sec^{-1}\left(\frac{x+1}{x-1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right)$, $x \in [0, \infty)$ and $x \neq 1$, then $\frac{dy}{dx}$ is equal to
 (a) 1 (b) $\frac{x-1}{x+1}$ (c) 0 (d) $\frac{x+1}{x-1}$

ANALYTICAL REASONING

41. You are given two (unmarked) containers of capacity 9 and 4 L and a huge tank of water. Need is to get a measure of exactly 6 L of water. A move is either filling a container completely or emptying a container (either fully or partially). The smallest number of moves needed to do this task is
 (a) 8 (b) 10 (c) 12 (d) None
42. What is the next letter in the series OTTFFSSSSN?
 (a) T (b) O (c) E (d) N
43. What is the diameter of the largest circle that can be drawn on a chessboard so that its entire circumference gets covered by the black squares and no part of the circumference on any white space, given that the chessboard has black and white squares of size one inch.
 (a) 1 inch (b) $\sqrt{2}$ inch
 (c) $\sqrt{10}$ inch (d) $2\sqrt{3}$ inch
44. A car is filled with $4\frac{1}{2}$ L of fuel for a around trip. If amount of fuel taken while going is $\frac{1}{4}$ th more than the amount taken for coming, what is the amount of fuel consumed when coming back?
 (a) 1.5 (b) 2 (c) 1.75 (d) None
45. Which of the following are greater than x when $x = \frac{9}{11}$?
 I. $\frac{1}{x}$ II. $\frac{x+1}{x}$ III. $\frac{x+1}{x-1}$
 (a) I only (b) I and II only
 (c) I and III only (d) II and III only
46. Four friends-Arjan, Bhuvan, Guran and Lakha comparing the number of sheep that they owned. It was found that Guran had ten more sheep than Lakha. If Arjan gave one-third to Bhuvan and Bhuvan gave a quarter of what he then held to Guran, who then passed on a fifth of his holding to Lakha, they would all have an equal number of sheep. How many sheep did each of them possess? Give the minimal possible answer.
 (a) 200, 105, 110, 100 (b) 90, 55, 55, 45
 (c) 180, 110, 110, 100 (d) None of these



47. In a class, six students P, Q, R, S, T and U are the top six rank holders, not necessarily in the same order. R did not get the 4th rank. P's rank is higher than U's and R's but lower than Q's. Among these six rankers, there are four students whose ranks are lower than S's rank and five students whose ranks are above that of T. who is ranked 5th in the class?

(a)U (b)T (c)R (d)None

48. Three players-Aalu, Kachaalu and Bhalu were playing poocker and suddenly started to quarrel among themselves blaming each other for cheating. It was found out that atleast one person among the three cheated. When they were asked who cheated, their replies were as follows.

Aalu I did not cheat, Kachaalu cheated

Kachaalu I did not cheat, Both Aalu and Bhalu cheated.

Bhalu I did not cheat, only Kachaalu did not cheat.

If exactly one person among them always spoke truth, another always lied and the third alternated between the truth and lie, then which of the following statements can never be true in any case?

- (a) Only Aalu and Bhalu cheated
(b) Only Aalu and Bhalu did not cheat
(c) Bhalu always spoke the truth
(d) Bhalu alternated between truth and lie.

49. If x and y are the two digits of the number 565xy such that this number is divisible by 80, then $x + y$ is equal to

(a)2 (b)3 (c)8 (d)6

50. If both 7^2 and 3^3 are factors of the number $(a \cdot 11^3 \cdot 6^2 \cdot 3^{11})$, then what is the smallest possible value of a?

(a)1323 (b)147 (c)21 (d)3087

51. Let x, y and z be distinct integers, x and y are odd and positive, and z is even and positive. Which one of the following statements cannot true?

- (a) $(x - z)^2 y$ is even (b) $(x - z) y^2$ is odd
(c) $(x - z) y$ is odd (d) $(x - y)^2 z$ is even

52. From a height of 16 m, a ball fell down and each time it bounces half the distance back. What is the distance travelled?

(a)45 m (b) ∞ (c)48 m (d)24 m

53. If a man walks at the rate of 4 km/h, he misses a train by only 6 min. However, if he walks at the rate of 5 km/h he reaches the station 6 min before the arrival of the train. Find the distance covered by him to reach the station.

(a)4 km (b)7 km (c)9 km (d)5 km

Direction (Q. Nos. 54-57) Read the following statements and answer questions.

The office staff of XYZ corporation presently consists of three book keepers, P, Q, R and 5 secretaries S, T, U, V, W. the management is planning to open a new office in another city using 2 book keepers and 3 secretaries of the present staff. To do so they plan to separate certain individuals who don't function well together. The following guidelines were established to set up the new office.

- (i) Book keepers P and R are constantly finding fault with one another and should not be sent together to the new office as a team.
(ii) R and T function well alone but as a team, they should be separated.
(iii) S and V have not been on speaking terms and shouldn't go together.
(iv) Since, S and U have been competing for promotion they shouldn't be a team
54. If P is to be moved as one of the bookkeepers, which of the the following cannot be a possible working unit?
(a)PQSTW (b)PQSVW
(c)PQTUW (d)PQTVW
55. If R and U are moved to the new office, how many combinations are possible?
(a)1 (b)2 (c)3 (d)4
56. If R is sent to the new office, which member of the staff cannot go with R?
(a)Q (b)S (c)W (d)V
57. If S goes to the new office, which of the following is true?
(a)Only R cannot go (b)Only P cannot go
(c)Only P and R cannot go (d)R cannot go and W must go

58. Substitute digits for the letters to make the following relation true.

S T I L L
+ W I T H I N

L I M I T S

Note: The leftmost letter can't be zero in any word. Also, there must be a one-to-one mapping between digits and letters. e.g., if you substitute 3 for the letters S, no other letter can be 3 and all other S in the puzzle must be 3.

- (a)98533 + 258056 = 356589
(b)41211 + 527013 = 938224
(c)98533 + 158056 = 256589
(d)47166 + 517013 = 614179
59. 12 members were present at a board meeting. Each member shook hands will all of the other memebbers before and after the meeting. How many hand shakes were there?
(a)118 (b)127 (c)132 (d)264
60. The letters P, Q, R, S, T, U and V not necessarily in that order represents seven consecutive integers from 22 to 33. U is as much less than Q as R is greater than S. V is greater than U. Q is the middle term. P is 3 greater than S. Can you find the sequence of letters from the lowest. Value to the highest value?
(a)PVSQRTU (b)SUTQPRV
(c)USVQPRV (d)TUSQPRV
61. There were a total of 10 bicyles and tricycles. If the total number of wheels was 24, how many tricycles were there?
(a)4 (b)6 (c)8 (d)2



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62. A person travels on a cycle from home to church on a straight road with wind against him. He took 4 h to reach there. On the way back to the home, he took 3 h to reach as wind was in the same direction. If there is no wind, how much times does he take to travel from home to church?

(a) 3 h 35 min 12 s (b) 3 h 32 min 32 s
(c) 3 h 30 min 00 s (d) 3 h 25 min 42 s

63. What are the next three numbers in the given series.

1 1 2 1 2 2 3 1 2 2 3 2 3 3 4 1 2 2 3 2 3 3 4 2 3 3?

(a) 2, 3, 4 (b) 2, 3, 2 (c) 1, 2, 3 (d) 4, 3, 4

64. In the middle of the confounded desert, there is the lost city of "Ash". To reach it, I will have to travel overland by foot from the coast. On a trek like this, each person can only carry enough rations for five days and the farthest we can travel in one day is 30 miles. Also, the city is 120 miles from the starting point. What I am trying to figure out is the fewest number of persons, including myself, that I will need in our group so that I can reach the city, stay overnight, and then return to the coast, without running out of supplies. How many persons (including myself) will I need to accomplish this mission?

(a) 5 (b) 6 (c) 4 (d) 3

65. A woman took a certain number of eggs to the market and sold some of them. The next day through her poultry industry, the number left over had been doubled, and she sold the same number as the previous day. On the third day, the new remainder was tripled, and she sold the same number as before. On the fourth day, the remainder was quadrupled, and her sale were the same as before. On the fifth day, what had been left over were quintupled, yet she sold exactly the same as on all the previous occasions and so disposed of her entire stock. What is the smallest number of eggs she could have taken to the market the first day, and how many did she sell daily?

(a) 110.50 (b) 127.65
(c) 100.60 (d) 103.60

66. The Bulls, Pacers, Lakers and Jazz ran for a contest. Anup, Sujit, John made the following statements regarding results.

- Anup said either Bulls or Jazz will definitely win.
- Sujit said he is confident that Bulls will not win.
- John said he is confident that neither Jazz nor Lakers will win.

When the results came, it was found that only one of the above three had made a correct statement. Who has made the correct statement and who has won the contest?

(a) Anup, Bulls (b) John, Pacers
(c) Sujit, Lakers (d) Sujit, Jazz

67. A certain street has 1000 buildings. A sign-maker is contracted to number the houses from 1 to 1000. How many zeroes will be needed?

(a) 128 (b) 190 (c) 181 (d) 192

68. Examine the following sequence of numbers

1

1 1

2 1

1 2 1 1

1 1 1 2 2 1

3 1 2 2 1 1

1 3 1 1 2 2 2 1

1 1 1 3 2 1 3 2 1 1

3 1 1 3 1 2 1 1 1 3 1 2 2 1

What are the next two numbers in the given series?

(a) 13211311122111231131 and

11213211321222111131221133

(b) 23113112211132113111 and

11121321132122211131221133

(c) 11231131122111321131 and

11131221212221133112132113

(d) 13211311123113112211 and

11131221133112132113212221

69. There were two men standing on a street. The one says to the other. "I have 3 daughters, the product of their ages is 36. What is the age of the OLDEST daughter?"

The second guy says, "I need more information." "So, the first guy says, "The sum of their ages is equal to the address of the house across the street."

The second guy looks at the address and says, "I still need more information". So, the first guy says. "My oldest daughter wears a red dress."

(a) 9 (b) 6 (c) 12 (d) 4

70. Three Gold (G) coins, three Silver (S) coins and three Copper (C) coins are arranged in a single row as follows

- Only 2 adjacent unlike coins can be moved at any one time.
- The moved coins must be in contact with atleast one other coin in line i.e., no pair of coins is to be moved and placed away from the remaining ones.
- No coin pairs can be reversed i.e., S-C combination must remain in that order in its new positions when it is moved.

What is the minimum number of moves required to get all the coins in following order?

C C C S S S G G G

(a) 6 (b) 9 (c) 8 (d) 12

71. Mr. and Mrs. Birla and Mr. and Mrs. Tata competed in a chess tournament. Of the three games played

(I) In only the first game were the two players married to each other.

(II) The men won two games and the women won one game.

(III) The Birlas won more games than the Tatas.

(IV) Anyone who lost a game did not play a subsequent game.

Who did not lose a game?

(a) Mr Birla (b) Mrs Birla
(c) Mr Tata (d) Mrs Tata





72. Of the three numbers, second is twice the first and is also thrice the third. If the average of three numbers is 44, the largest number is
(a)24 (b)36 (c)72 (d)108
73. Larger, medium and small ships are used to bring water. 4 large ships carry as much water as 7 small ships. 3 medium ships carry the same amount of water as 2 larger ships and 1 small ship. 15 large, 7 medium and 14 small ships, each made 36 journeys and brought a certain quantity of water. In how many journeys would 12 large, 14 medium and 21 small ships bring the same quantity of water?
(a)32 (b)25 (c)29 (d)49
74. Five men P, Q, R, S and T read newspaper. The one who reads first gives it to R. the one who reads last had taken it from P. T was not the first or the last to read. There were two readers between Q and P. To whom did Q pass the newspaper?
(a)R (b)P (c)S (d)T
75. An airline has a certain free luggage allowance and charges for excess luggage at a fixed rate per kg. Two passengers Raja and Rahim have 60 kg luggage between them, and are charged ₹ 1200 and ₹ 2400, respectively for excess luggage. Had the entire luggage belonged to one of them, the excess luggage charge would have been ₹ 5400. What is the weight of Rahim's luggage?
(a)20 kg (b)25 kg (c)30 kg (d)35 kg
76. A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. What number of rows is not possible?
(a)3 (b)4 (c)5 (d)6

Directions (Q. Nos. 77-80) Read the following passage and answer the questions.

Sports (and game) persons P, Q, R, S, T, U and V of a university are at the Bengaluru Airport. Five of them are selected players and leaving to participate in the grand sports Event in five different events cricket, carrom, badminton and table tennis being held at 5 different cities Mumbai, Chennai, Kolkata, Delhi and Hyderabad.

- (i) P is going to Delhi, but he does not play either cricket or carrom
(ii) Q has come to give send off to R, who is a chess player and is not leaving to either Mumbai or Hyderabad.
(iii) S is leaving to Kolkata to play table tennis.
(iv) U is leaving to Mumbai but he different does not play either badminton or cricket.
(v) T is not a selected player.

77. Who plays badminton?
(a)P (b)Q
(c)R (d)S
78. Cricketer goes to
(a)Mumbai (b)Hyderabad
(c)Chennai (d)Delhi

79. Player of which game goes to Delhi?
(a)Badminton (b)Chess
(c)Cricket (d)Table tennis
80. Who plays chess and where is he going?
(a)R and Chennai (b)S and Mumbai
(c)U and Delhi (d)None of these

Directions (Q. Nos. 81-85) Read the following information carefully and answer the questions.

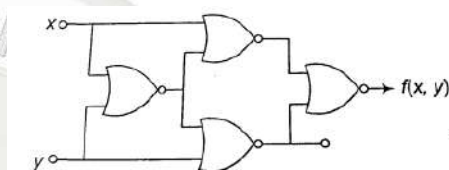
- (i) $P \psi Q$ means P is mother of Q
(ii) $P \in Q$ means P is sister of Q
(iii) $P \$ Q$ means P is father of Q
(iv) $P \# Q$ means P is brother of Q
81. Which of the following means N is definitely daughter of K?
(a) $K \$ L \# M \# N$ (b) $M \psi K \$ N \in L$
(c) $K \psi M \# L \in N$ (d) $L \psi K \$ N \# M$
82. Which of the following means R is brother of T?
(a) $R \psi S \# U \$ T$ (b) $U \psi R \# S \# T$
(c) $U \psi R \in S \psi T$ (d) $T \# \$ \$ Q \in R$
83. Which of the following means X is real grandmother of Y?
(a) $X \in Z \psi K \$ L \# Y$ (b) $Y \psi K \$ X \# L$
(c) $Y \# L \$ K \psi X \in Z$ (d) $K \# X \psi Z \# L \$ Y$
84. If $K \psi L \in M \# N$, then how K is related with N?
(a)Mother (b)Aunt
(c)Great Aunt (d)Grandmother
85. Which of the following means K is related K is nephew of M?
(a) $N \# M \$ L \# K \in O$ (b) $K \# L \$ N \in O \$ M$
(c) $L \psi O \# M \$ O \in K$ (d) $M \# N \$ L \# K \$ O$
86. There are six houses in a row. Mr. Lal has Mr. Babu and Mr. Anil as neighbours. Mr. Bhatia has Mr. Gupta and Mr. Sharma as neighbours. Mr. Gupta's houses is no next to Mr. Babu or Mr. Anil and Mr. Sharma does not live next to Mr. Anil. Who are Mr. Babu's next-door neighbours?
(a)Mr. Lal and Mr. Bhatia (b)Mr. Lal and Mr. Anil
(c)Mr. Sharma and Mr. Lal (d)Only Mr. Lal
87. A watch which gains 10 s in 5 min was set correct at 9 am when the watch indicated 20 min past 7 O' clock, the same evening, the true time is
(a)7 pm (b)7 : 40 pm (c)7 : 10 pm (d)8 pm
88. A boy observes the reflection of a clock in a mirror. The time observed by the boy in the mirror is 3 h 45 min. what is the actual time shown in the clock?
(a)8 h 45 min (b)9 h 45 min
(c)8 h 15 min (d)9 h 15 min
89. Gold is 19 times as heavy as water and copper is 9 times as heavy as water. In what ratio should these be mixed to get an alloy 15 times as heavy as water?
(a)1 : 2 (b)2 : 3 (c)3 : 2 (d)1 : 1
90. In an objective type examination, 120 objective type questions are there : each with 4 options P, Q, R and S. A candidate can choose either one of these options or can leave the question unanswered. How many different ways exist for answering this question paper?
(a) 5^{120} (b) 4^{120} (c) 120^5 (d) 120^4



COMPUTER AWARENESS

91. A CPU has an arithmetic unit that adds bytes and then sets its V, C and Z flag bits as follows: the V-bit is set if arithmetic overflow occurs. The C-bit is set if a carry-out is generated from the most significant bit during an operation. The Z-bit is set if the result is zero. What are the values of the V, C and Z flag bits respectively after the 8-bit bytes 1100 1100 and 1000 1111 are added?
- (a) 0, 0, 0 (b) 1, 1, 0
(c) 1, 1, 1 (d) 0, 1, 0
92. Which of the following statements is always true?
- (a) A compiled program uses more memory than an interpreted program.
(b) A compiler converts a program to a lower level language for execution.
(c) A compiler for a high level language takes less memory than its interpreter
(d) Compiled programs take more time to execute than interpreted programs.
93. Floating point numbers in a computer are represented using a 10-bit mantissa (including a sign bit) and a 7-bit exponent (including a sign bit). What is the approximate value of the maximum number, which can be represented? Assume that the mantissa is stored in the normalised form, that is, without leading zeroes.
- (a) 2^{128} (b) 2^{127} (c) 2^{64} (d) 2^{63}
94. The capacity of a memory unit is defined by the number of words multiplied by the number of bits per word. How many separate address and data line are needed for a memory of $4K \times 16$?
- (a) 10 address lines and 16 data lines
(b) 12 address lines and 10 data lines
(c) 12 address lines and 16 data lines
(d) 12 address lines and 8 data lines
95. The main disadvantage of direct mapping of cache organization is that
- (a) it doesn't allow simultaneous access to the intended data and its tag
(b) it is more expensive than other type of organizations.
(c) the cache hit ratio is degraded if two or more blocks used alternatively map onto the same block frame in the cache.
(d) the number of blocks required for the cache increases linearly with the size of the main memory.
96. Let $A[1 \dots 10]$ be an array, let $A[i] = 2i$ for $1 \leq i \leq 10$. After the assignment $j = A[A[5]]$ is executed, the value of $A[j]$ is equal to
- (a) undefined (b) 1
(c) 5 (d) 10
97. The first instruction of bootstrap loader program of an operating system is stored in
- (a) RAM (b) BIOS

- (c) Hard disk (d) None of these
98. The function $AB'C + AB'C + ABC' + A'B'C$ is equivalent to
- (a) $AC' + AB + A'C$ (b) $AB' + AC' + A'C$
(c) $A'B + AC' + AB'$ (d) $A'B + AC + AB'$
99. The addition of 4 bit, 2's complement binary numbers 1101 and 0100 results is
- (a) 0001 and an overflow (b) 1001 and no overflow
(c) 0001 and no overflow (d) 1001 and an overflow
100. Given $\sqrt{(224)}_r = (13)_r$, the value of radix r is
- (a) 10 (b) 8 (c) 6 (d) 5
101. Let $A = 11111010$ and $B = 00001010$ be two 8 bit 2's complement number. Their product in 2's complement is
- (a) 11000100 (b) 10011100
(c) 10100101 (d) 11010101
102. Identify the logic function performed by the circuit



- (a) Exclusive OR (b) Exclusive NOR
(c) NAND (d) NOR
103. Which of the following is (are) true about virtual memory system that uses pages?
- (I) The virtual address space can be larger than the amount of physical memory.
(II) Programs must be resident in main memory throughout their execution.
(III) Pages correspond to semantic characteristics of the programs.
- (a) I only (b) II only
(c) I and II (d) I and III
104. The minimum number of gates needed to implement the Boolean function $f(x, y, z) = z(x + y) + (\bar{z} + x + y)(\bar{x} + \bar{y})$ is
- (a) 2 (b) 3 (c) 4 (d) 5
105. How many bits are required to store an ASCII character?
- (a) 7 (b) 6 (c) 8 (d) None

GENERAL ENGLISH

Direction (Q. Nos. 106-107) Read the following information carefully and answer the questions.

Each sentence given in the questions has two blanks, each blank indicating that something has been omitted. Beneath the sentence are four sets of words. Choose the set of words for each blank that best fits the meaning of the sentence as a whole.

106. Greek philosophers tried to contemporary notions of change and stability by postulating the existence of the atom, particle from which all varieties of matter are formed.
- (a) confirm... an interesting (b) reconcile..... an indivisible
(c) simplify... a specific (d) eliminate... an infinitesimal



107. The Tata group will need all its considerable management And to manage tough challenges ahead after taking over Corus Steel.

- (a) skills...interests (b) knowledge...manpower
(c) acumen...onus (d) experience...brand equity

Direction (Q. Nos. 108-109) Read the following information carefully and answer the questions.

In each of the following questions, a related pair of words or phrases is followed by four pairs of words or phrases. Select the pair that best expresses a relationship similar to that expressed in the original pair.

108. INFLAMMABLE : IGNITED :: :

- (a) Fragile : Shattered (b) Flexible : Broken
(c) Famous : Plagiarized (d) Somber : Mourned

109. SAVANT : OBTUSE :: :

- (a) Seer : Ominous (b) Writer : Verbose
(c) Judge : Melodramatic (d) Athlete : Sluggish

Direction (Q. Nos. 110-111) Each question consists of a word printed in capital letters, followed by four words or phrases. Choose the word or phrase that is **most nearly opposite** in meaning to the word in capital letters.

110. OPPROBRIUM

- (a) Honour (b) Prudence
(c) Ostentation (d) Umbrage

111. INCESSANT

- (a) Perpetual (b) Persistent
(c) Sporadic (d) Unrelenting

Direction (Q. Nos. 112-113) Each question consists of a word printed in capital letters, followed by four words or phrases. Choose the word or phrase that is **most similar** in meaning to the word in capital letters.

112. EXASPERATE

- (a) Pacify (b) Mollify
(c) Irritate (d) Placate

113. INIMICAL

- (a) Antagonistic (b) Anonymous
(c) Fanciful (d) Accurate

Direction (Q. Nos. 114-116) Read the following passage and answer the questions, based on what is stated or implied in the passage.

Declassification of government documents has shed new light on the events comprising the Cuban missile Crisis of October 1962. Prior to the accessibility of these records, the only source of account to the Crisis for scholars and historians were the personal memoirs and narratives of the officials who served under Kennedy and Krushchev during this period. Many of the declassified documents are transcriptions and notes of meetings between members of the CIA and President Kennedy's Cabinet, as well as the President himself. The revelations in these documents have demonstrated the inadvertent inaccuracies and intended obscurities inherent in the first-person narratives of the Crisis, and has aided historians from all three countries involved in the Crisis to get

a more authentic representation of what truly transpired, and for what reasons. Of perhaps the most interest to historians are declassified correspondence between John F. Kennedy and Nikita Krushchev that challenge the idea that the height of the Crisis extended only over the course of thirteen days. Indeed, these letters indicate that the Crisis was far from resolved by Krushchev's October 28 decision to withdraw the Soviet missiles from Cuba ; instead it endured far into the following month, while America slept fitfully under the illusion of peace.

114. The author is mainly concerned with

- (a) Petitioning the government to make all classified documents of historic interest accessible to the general public
(b) Discounting the sense of danger many Americans felt during the Cuban Missile Crisis
(c) Revealing a calculated deception perpetrated by members of Kennedy's Cabinet.
(d) Illustrating how previously accepted ideas based on hearsay are being refuted by concrete evidence.

115. According to the passage, which of the following statement(s) is/are true of the Cuban Missile Crisis ?

- (I) The Crisis is still shrouded in mystery
(II) The memoirs of those closely involved in the Crisis were not entirely factual
(III) The Crisis spanned thirteen tense days
(a) I only (b) II only
(c) III only (d) II and III only

116. The author's use of the phrase "inadvertent inaccuracies and intended obscurities" suggests all of the following EXCEPT

- (a) historical record is often skewed by human perception
(b) details of the Crisis were purposely omitted or vague
(c) every politician deals in deception and prevarication
(d) memory is incapable of recapturing the full details of an event

Directions (Q. Nos. 117 -118) In each of the following questions, a sentence is given with a blank followed by four alternatives. Choose the word or phrase that most correctly completes the sentences.

117. Mary did not attend office yesterday. She for a picnic.

- (a) will have gone (b) have gone
(c) may have gone (d) would go

118. I don't know where Maya is. She at home.

- (a) would be (b) is
(c) can be (d) could be

119. Choose the most appropriate meaning for the following idiom "To fish in troubled waters"

- (a) To make the situation worse
(b) To make profit when others are in trouble
(c) To create trouble for others
(d) To indulge in evil acts

120. Read the following sentence and choose one underlined word or phrase that would not be appropriate in standard English. One of the chair's leg was broken and the upholstery needed mending

- (a) the (b) chair's (c) legs (d) needed





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PYQ No. : 02

ANSWER KEY

1. (b)	21. (c)	41. (a)	61. (a)	81. (b)	101. (a)
2. (c)	22. (d)	42. (d)	62. (d)	82. (b)	102. (b)
3. (d)	23. (c)	43. (b)	63. (d)	83. (d)	103. (c)
4. (a)	24. (b)	44. (b)	64. (c)	84. (a)	104. (a)
5. (b)	25. (a)	45. (b)	65. (d)	85. (d)	105. (a)
6. (b)	26. (d)	46. (d)	66. (c)	86. (c)	106. (b)
7. (b)	27. (c)	47. (c)	67. (d)	87. (a)	107. (c)
8. (a)	28. (c)	48. (d)	68. (a)	88. (c)	108. (d)
9. (b)	29. (b)	49. (d)	69. (a)	89. (c)	109. (d)
10. (d)	30. (b)	50. (b)	70. (a)	90. (a)	110. (a)
11. (a)	31. (b)	51. (a)	71. (a)	91. (b)	111. (c)
12. (b)	32. (c)	52. (c)	72. (c)	92. (b)	112. (c)
13. (c)	33. (b)	53. (a)	73. (c)	93. (d)	113. (a)
14. (a)	34. (c)	54. (b)	74. (a)	94. (c)	114. (d)
15. (d)	35. (c)	55. (a)	75. (d)	95. (d)	115. (d)
16. (c)	36. (c)	56. (b)	76. (d)	96. (a)	116. (c)
17. (a)	37. (c)	57. (d)	77. (a)	97. (b)	117. (b)
18. (c)	38. (b)	58. (a)	78. (b)	98. (b)	118. (a)
19. (b)	39. (a)	59. (c)	79. (a)	99. (c)	119. (b)
20. (b)	40. (c)	60. (d)	80. (a)	100. (d)	120. (c)

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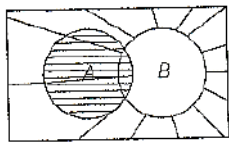


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MATH SOLUTION

1. (b)



$$P(A') = 0.3, P(B) = 0.5 \text{ and } P(A \cap B) = 0.3$$

$$P(A) = 1 - P(A') = 1 - 0.3 = 0.7$$

$$P\left(\frac{B}{A \cup B'}\right) = \frac{P(B \cap (A \cup B'))}{P(A \cup B')} = \frac{P(A \cap B)}{P(A \cup B)}$$

$$P(A \cup B) = 1 - P(B') + P(A \cap B)$$

$$0.3 = \frac{0.3}{1 - 0.5 + 0.3} = \frac{0.3}{0.8} = \frac{3}{8}$$

2. (c)

Given lines,

$$x^2(\tan^2 \theta + \cos^2 \theta) + 2xy \tan \theta - y^2 \sin^2 \theta = 0$$

Its angle bisector $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

$$\frac{x^2 - y^2}{\tan^2 \theta + \cos^2 \theta - (-\sin^2 \theta)} = \frac{xy}{\tan \theta}$$

$$\Rightarrow \frac{x^2 - y^2}{\sec^2 \theta} = \frac{xy}{\tan \theta}$$

$$\text{At } \theta = \frac{\pi}{3}, \frac{x^2 - y^2}{4} = \frac{xy}{\sqrt{3}}$$

Also, $y = mx$ satisfies Eq. (i)

$$\frac{x^2 - m^2 x^2}{4} = \frac{x^2 m}{\sqrt{3}}$$

$$\Rightarrow \sqrt{3}(1 - m^2) = 4m$$

$$\Rightarrow \sqrt{3}m^2 + 4m = \sqrt{3}$$

3. $I = \int_{-\pi/2}^{\pi/2} [f(x) + f(-x)][g(x) - g(-x)] dx \dots (1)$

$$I = \int_{-\pi/2}^{\pi/2} [f(x) + f(x)][g(x) - g(x)] dx$$

$$I = - \int_{-\pi/2}^{\pi/2} [f(-x) + f(x)][g(x) - g(-x)] dx \dots (2)$$

$$\int_{-a}^a f(x) dx = \int_{-a}^a f(-x) dx$$

$$= 2 \int_{-a}^a f(x) dx$$

$$= 2I = 0$$

Adding equation (1) and (2)

$$2I = 0$$

$I = 0$ option (d)

4. $\cot \alpha_1 \cdot \cot \alpha_2 \cdot \cot \alpha_3 \dots \cot \alpha_n = 1$

Split in $\frac{\cos \alpha}{\sin \alpha}$

$$\cos \alpha_1 \cdot \cos \alpha_2 \cdot \cos \alpha_3 \dots \cos \alpha_n = \sin \alpha_1 \sin \alpha_2 \dots \sin \alpha_n$$

$$\text{multiplying } (\cos \alpha_1 \cdot \cos \alpha_2 \cdot \dots \cos \alpha_n)^2 = \sin \alpha_1 \cdot \cos \alpha_1 \cdot \sin \alpha_2 \cdot$$

$$\cos \alpha_2 \cdot \dots \sin \alpha_n \cdot \cos \alpha_n$$

divide by 1 and 2 multiply right side

$$(\cos \alpha_1 \cdot \cos \alpha_2 \cdot \dots \cos \alpha_n)^2 = \frac{1}{2^n} (\sin 2\alpha_1)(\sin 2\alpha_2) \dots (\sin 2\alpha_n)$$

$$(\cos \alpha_1 \cdot \cos \alpha_2 \cdot \dots \cos \alpha_n)^2 \leq \frac{1}{2^n}$$

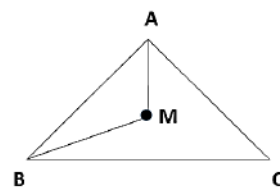
$$(\cos \alpha_1 \cdot \cos \alpha_2 \cdot \dots \cos \alpha_n) < \frac{1}{2^{n/2}}$$

Option (d)

5. Let M is inside triangle $\triangle ABC$

In $\triangle ABM$

$$AB < AM + MB \dots (i)$$



In $\triangle ACM$

$$AC < AM + MC \dots (ii)$$

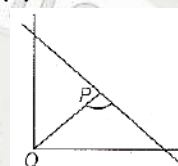
Adding (i) + (ii)

$$AB + AC < 2AM + MB + MC$$

$$[AB + AC < MB + MC] \text{ since M inside } \triangle ABC$$

$$1 AM < AB + BC$$

6. (b) Before rotation of axes



$$\text{Equation of lines, } \frac{x}{a} + \frac{y}{b} = 1 \dots (i)$$

After rotation,

$$\text{Equation of lines, } \frac{x}{p} + \frac{y}{q} = 1 \dots (ii)$$

But in both case perpendicular distance from origin will remains same

$$\frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{1}{\sqrt{\frac{1}{p^2} + \frac{1}{q^2}}} \Rightarrow \frac{1}{p^2} + \frac{1}{q^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

7. (b) $a + b = -p$, $ab = 1$; $c + d = -q$; $cd = 1$

$$E = (a - c)(b - c)(a + d)(b + d)$$

$$= [(a - c)(b + d)] \cdot [(b - c)(a + d)]$$

$$= [ab + ad - cd - cd] \cdot [ab + bd - ca - cd]$$

$$= [1 + ad - cb - 1] [1 + bd - ca - cd]$$

$$= (ad - cb)(bd - ca)$$

$$= abd^2 - cd - cb^2 d + c^2 ab$$

Put $ab = 1$ and $cd = 1$

$$= d^2 - a^2 - b^2 + c^2 = (c^2 + d^2) - (a^2 + b^2)$$

$$= (c^2 + d^2 + 2cd - 2cd) - (a^2 + b^2 + 2ab - 2ab)$$

$$= [(c + d)^2 - 2] - [(a + b)^2 - 2] = (-q)^2 - (-p)^2$$

$$= q^2 - p^2$$

It has an alternate Method:

$$E = (a - c)(b - c)(a + d)(b + d)$$

$$= (ab - ac - bc + c^2)(ab + ad + bd + d^2)$$

As we know $ab = cd = 1$, Hence

$$E = (1 - c(a + b) + c^2)(1 + d(a + b) + d^2)$$

As we know $a + b = -p$, $c + d = -q$, Hence,

$$E = (1 - c(-p) + c^2)(1 + d(-p) + d^2)$$

$$E = (c^2 + pc + 1)(d^2 - dp + 1)$$

As we know c, d are roots of $x^2 + qx + 1 = 0$, hence, c, d satisfy given equation



$$c^2 + qc + 1 = 0 \Rightarrow c^2 + 1 = -qc$$

$$\text{and } d^2 + qd + 1 = 0 \Rightarrow d^2 + 1 = -qd$$

Put these values in Eq. (i), we get

$$E = (pc - qc)(-dp - qd) = -cd(p^2 - q^2)$$

$$= q^2 - p^2$$

8. (a) $f(x) + f(1-x) = 2$

$$f\left(\frac{1}{2001}\right) + f\left(\frac{2}{2001}\right) + \dots + f\left(\frac{2000}{2001}\right)$$

$$f(x) + f(1-x) = 2$$

$$f\left(\frac{1}{2001}\right) + f\left(1 - \frac{1}{2001}\right) = 2 \Rightarrow f\left(\frac{1}{2001}\right) + f\left(\frac{2000}{2001}\right) = 2$$

$$f\left(\frac{2}{2001}\right) + f\left(1 - \frac{2}{2001}\right) = 2 \Rightarrow f\left(\frac{2}{2001}\right) + f\left(\frac{1999}{2001}\right) = 2$$

$$f\left(\frac{1000}{2001}\right) + f\left(1 - \frac{1000}{2001}\right) = 2 \Rightarrow f\left(\frac{1000}{2001}\right) + f\left(\frac{1001}{2001}\right) = 2$$

Total these types of pairs are 1000.

Hence, answer is $1000 \times 2 = 2000$

9. (b) Given, $e^{1/c}, e^{b/ac}, e^{1/a}$
 a, b, c are in AP $\Rightarrow 2b = c + a$
 In case of exponential, first we assume that it will be in G.P.

$$\Rightarrow e^{1/c} \times e^{1/a} = (e^{b/ac})^2$$

$$\Rightarrow e^{\frac{1}{c} + \frac{1}{a}} = e^{\frac{2b}{ac}}$$

$$\Rightarrow e^{\frac{a+c}{ac}} = e^{\frac{2b}{ac}} \text{ as, given } 2b = a + c$$

$$\Rightarrow e^{\frac{2b}{ac}} = e^{\frac{2b}{ac}}$$

$$\therefore e^{1/c}, e^{b/ac}, e^{1/a} \text{ in GP.}$$

10.(d) $x^2 + x + 1 = 0$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1-4(1)(1)}}{2(1)} = \frac{-1 \pm \sqrt{3}i}{2} \Rightarrow x = \frac{-1 \pm \sqrt{3}i}{2}$$

$$\Rightarrow x = \omega, \omega^2$$

ω, ω^2 are cube roots of unity

$$\omega^3 = 1$$

$$\text{Let } \alpha = \omega \text{ and } \beta = \omega^2$$

Given root of Eq. to find

$$\Rightarrow \alpha^{19}, \beta^7$$

$$\alpha^{19} = \omega^{19} = (\omega^3)^6 \cdot \omega = \omega(1) = \omega = \alpha \quad [\because \omega^3 = 1]$$

$$\beta^7 = (\omega^2)^7 = \omega^{14} = (\omega^3)^4 \cdot \omega^2 = \omega^2 = \beta$$

$$\Rightarrow \alpha^{19} = \alpha \text{ and } \beta^7 = \beta$$

Hence, new equation will remain same.

11. (a) Let $f(x) = (x+1)(x+4)(x+9) \dots (x+400)$

$$\Rightarrow (x+1^2)(x+2^2)(x+3^2) \dots (x+20^2)$$

Put $f(x) = 0$, we get

$$x = -1^2, -2^2, -3^2, -4^2, \dots, -20^2 \text{ roots of } f(x)$$

Highest power of x in $f(x)$ is x^{20}

Using Vieta's theorem

$$f(x) = x^{20} - (\text{sum of roots taken one at a time})$$

$$x^{19} + (\text{sum of root taken two at a time}) x^{18} + \dots \text{sum of roots taken one at a time}$$

$$= \sum \alpha = \alpha_1 + \alpha_2 + \alpha_3 + \dots$$

$$= -1^2 - 2^3 = 3^2 - 4^2 - \dots - 20^2 x^{19} + [(-1)^2(-2)^2 + (-2)^3(-3)^2 + (-3)^2(-4)^2 + \dots] x^{18} + \dots$$

$$\text{Coefficient of } x^{19} = (1^2 + 2^2 + 3^2 + 4^2 + \dots + 20^2)$$

$$= \frac{20(21)(41)}{6}. \text{ Put } (n=20)$$

$$= 2870$$

12. (b) $y = 0.36 \log_{\frac{25}{100}} \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \right)$

$$y = 0.36 \log_{\frac{1}{4}} \left(\frac{\frac{1}{3}}{1 - \frac{1}{3}} \right) = 0.36 \log_{2^{-2}} (2^{-1})$$

$$y = 0.36 \log_{2^{-1}} 2^{-1} = \frac{0.36}{2} \log_2 2 = 0.4 \log_2 2$$

$$y = 0.18$$

13. (c) $H_1, H_2, \dots, H_N, b$ in HP

14. (a) $a > 0, a \neq 1, x > 0, x \neq 1$

$$\Rightarrow \frac{2}{\log_a x} + \frac{1}{\log_a x} + \frac{3}{\log_a a^2 + \log_a x} = 0$$

$$\Rightarrow \frac{2}{\log_a x} + \frac{1}{\log_a a + \log_a x} + \frac{3}{\log_a a^2 + \log_a x} = 0$$

$$\Rightarrow \frac{2}{\log_a x} + \frac{1}{1 + \log_a x} + \frac{3}{2 + \log_a x} = 0$$

$$\text{Let } \log_a x = t$$

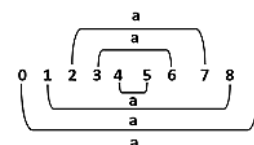
$$\Rightarrow \frac{2}{t} + \frac{1}{1+t} + \frac{3}{2+t} = 0$$

$$\Rightarrow \frac{2(1+t)(2+t) + t(2+t) + 3(1+t)t}{t(1+t)(2+t)} = 0$$

$$\Rightarrow 6t^2 + 11t + 4 = 0 \Rightarrow t = -\frac{1}{2}, -\frac{4}{3}$$

So, it has 2 solutions.

15. (d)



$$7 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 7 \times 7!$$

$$= 7 \times 7!$$

$$= 7 \times 7!$$

$$= 7 \times 7!$$

$$= 8!$$

$$\text{Total no. of war} = 4 \times 7 \times 7! + 8!$$

$$= 28 \times 7! + 8 \times 7!$$

$$= 36 (7!)$$





16. (c) Let $l = 7^m + 7^n$

7^r ends in 7, 9, 3, 1 according to r is the form of $4k + 1, 4k + 2, 4k - 1, 4k$

Sample space $n(s) = (100)^2$

$7^m + 7^n$ is divisible by 5, if

(i) m in form of $4k + 1$ and n is $4k - 1$

(ii) m is in form of $4k + 2$ and n is of the form $4k$ their vice versa.

Number of favourable order pairs

$$(m, n) = 2 \times 2 \times 25 \times 25$$

$$= 100 \times 25$$

$$(m, n) = 2500$$

17. (a) $x^3 - 3px^2 + 3qx - 1 = 0$

Its roots are a, b and c

Here we use Vieta's theorem

$$a + b + c = 3p, ab + bc + ac = 3q, abc = 1$$

$$\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$$

$$\text{Centroid } (x, y) = \left(\frac{a+b+c}{3}, \frac{1}{3}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)\right)$$

$$= \left(\frac{3p}{3}, \frac{1}{3}\left(\frac{ab+bc+ac}{abc}\right)\right)$$

$$= \left(p, \frac{1}{3} \times \frac{3q}{1}\right) = (p, q)$$

18. (c) Parabola $y^2 = 4x$ Circle $(x - 3)^2 + y^2 = 9$

$$\therefore a = 1$$

Center $(3, 0)$, Radius: $r = 3$

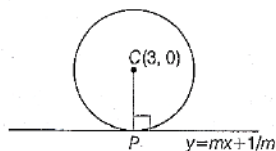
Let tangent of parabola be: $y = mx + \frac{a}{m}$ ($m = \text{slope}$)

$$y = mx + \frac{1}{m} (\because a = 1)$$

Also Eq. (i) is a tangent to the given circle

$$\therefore CP = r = 3$$

$$\left|\frac{3m - 0 + \frac{1}{m}}{\sqrt{m^2 + 1}}\right| = 3$$



$$\left(3m + \frac{1}{m}\right)^2 = 9(1 + m^2)$$

$$\Rightarrow 9m^2 + \frac{1}{m^2} + 6 = 9 + 9m^2$$

$$\Rightarrow m^2 = \frac{1}{3} \Rightarrow m = \pm \frac{1}{\sqrt{3}} \Rightarrow \sqrt{3}y = \pm(x + 3)$$

Now, we need a common tangent above X-axis.

$$\therefore y > 0$$

$$\therefore \sqrt{3}y = (x + 3)$$

19. (b) $|x^2 - x - 6| = x + 2$

$$\therefore |x| = a$$

$$\Rightarrow x = \pm a \text{ if } a \geq 0$$

$$\Rightarrow x^2 - x - 6 = \pm(x + 2) \text{ if } (x + 2) \geq 0 \Rightarrow x \geq -2$$

$$\Rightarrow x^2 - x - 6 = x + 2; x^2 - x - 6 = -x - 2$$

$$\Rightarrow x^2 - 2x - 8 = 0; x^2 - 4 = 0$$

$$\Rightarrow (x + 4)(x - 2) = 0; (x + 2)(x - 2) = 0$$

$$\Rightarrow x = 4, -2; x = 2, -2$$

$$\Rightarrow x = -2, 2, 4$$

20. (b)

When two dice are rolled, total no. of cases $= 6 \times 6 = 36$

Let A is an event when sum of 5 occurs $=$

$$\{(1, 4), (4, 1), (2, 3), (3, 2)\}$$

There are total 4 cases $P(A) = \frac{4}{36} = \frac{1}{9}$

Let B is an event sum 7 occurs

Total 6 cases

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

Let C be the event when neither 5 nor 7 as a sum occurs

$$= 1 - P(A) - P(B) = 1 - (1/9) - (1/6) = 26/36 = 13/18$$

Now, the probability that 5 comes before 7 as a sum.

$$P(A) + P(C) \times P(A) + P(C) \times P(C) + P(A) + \dots$$

$$= \frac{P(A)}{[1 - P(C)]} = \frac{1/9}{1 - 13/18} = \frac{2}{5}$$

21. (c) Letters of word STATISTICS are A IICSSSTTT

Letters of word ASSISTANT are AAINSSSTT

Common letters in both are A, I, S and T

$$\text{Probability of selecting A from both words} = \frac{1}{10} \times \frac{2}{9} = \frac{2}{90}$$

$$\text{Probability of selecting I from both words} = \frac{2}{10} \times \frac{1}{9} = \frac{2}{90}$$

$$\text{Probability of selecting S from both words} = \frac{3}{10} \times \frac{3}{9} = \frac{9}{90}$$

$$\text{Probability of selecting T from both words} = \frac{3}{10} \times \frac{2}{9} = \frac{6}{90}$$

$$\text{Probability of required event} = \frac{2}{90} + \frac{2}{90} + \frac{9}{90} + \frac{6}{90} = \frac{19}{90}$$

22. (d) Bag contains 6 red and 4 green balls.

If outcome on dice is 1 then only 1 ball can be draw and

$$\text{probability that it will be red} = \frac{1}{6} \times \frac{6}{10}$$

If outcome on dice is 2 then only 2 balls can be draw and

$$\text{probability that it will be red} = \frac{1}{6} \times \frac{{}^6C_2}{{}^{10}C_2}$$

If outcome on dice is 3 then only 3 balls can be draw and

$$\text{probability that it will be red} = \frac{1}{6} \times \frac{{}^6C_3}{{}^{10}C_3}$$

If outcome on dice is 4 then only 4 balls can be draw and

$$\text{probability that it will be red} = \frac{1}{6} \times \frac{{}^6C_4}{{}^{10}C_4}$$

If outcome on dice is 5 then only 5 balls can be draw and

$$\text{probability that it will be red} = \frac{1}{6} \times \frac{{}^6C_5}{{}^{10}C_5}$$

If outcome on dice is 6 then only 6 balls can be draw and

$$\text{probability that it will be red} = \frac{1}{6} \times \frac{{}^6C_6}{{}^{10}C_6}$$

Required probability

$$= \frac{1}{6} \times \left[\frac{{}^6C_1}{{}^{10}C_1} + \frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^6C_3}{{}^{10}C_3} + \frac{{}^6C_4}{{}^{10}C_4} + \frac{{}^6C_5}{{}^{10}C_5} + \frac{{}^6C_6}{{}^{10}C_6} \right] = \frac{1}{5}$$



23. (c) Volume of parallelepiped = $[a, b, c]$

$$\begin{vmatrix} 1 & \lambda & 1 \\ 0 & 1 & \lambda \\ \lambda & 0 & 1 \end{vmatrix} = \lambda^3 - \lambda - 1$$

$$f(\lambda) = \lambda^3 - \lambda - 1$$

$$\text{For minimum } f'(\lambda) = 3\lambda^2 - 1 = 0$$

$$\lambda^2 = \frac{1}{3}$$

$$\left(\lambda - \frac{1}{\sqrt{3}} \right) \left(\lambda + \frac{1}{\sqrt{3}} \right)$$

$$\text{So, clearly } \lambda = \frac{1}{\sqrt{3}}$$

24. (b) Let probability of getting odd number = p

$$\text{Then probability of getting even number} = \frac{p}{3}$$

In a dice, all outcomes getting $\{1, 2, 3, 4, 5, 6\}$ are mutually exclusive and exhaustive cases hence

$$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$$

$$p + p/3 + p + p/3 + p + p/3 = 1$$

$$\text{Hence, } p = 1/4$$

$$\text{So, } P(\text{odd number}) = p = 1/4,$$

$$P(\text{even number}) = p/3 = 1/12$$

$$P(\text{sum of number is even}) = P(\text{both number are even})$$

$$+ P(\text{both number are odd})$$

$$= \frac{3}{12} \times \frac{3}{12} + \frac{3}{4} \times \frac{3}{4}$$

$$= \frac{5}{8}$$

25. (a) Here we use Baye's theorem

Let E_1 denote the event that the letter came from

TATANAGAR

And E_2 denote the event that the letter came from

CALCUTTA.

Let A be the event that the two consecutive alphabets visible on the envelope are TA.

Since, the letters have come from either CALCUTTA or TATANAGER.

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$$

If E_1 has occurred then the letter has come from TATANAGAR. In the word TATANAGAR there are 8 consecutive alphabets

i.e. $\{CA, AL, LC, CU, UT, TT, TA\}$ and TA occurs once

$$P\left(\frac{A}{E_2}\right) = \frac{1}{7}$$

The probability that the letter has come from CALCUTTA is $P(E_2/A)$

By Baye's theorem,

$$\therefore P\left(\frac{E_2}{A}\right) = \frac{P(E_2)P\left(\frac{A}{E_2}\right)}{P(E_2)P\left(\frac{A}{E_2}\right) + P(E_1)P\left(\frac{A}{E_1}\right)}$$

$$= \frac{\frac{1}{2} \times \frac{1}{7}}{\frac{1}{2} \times \frac{1}{7} + \frac{1}{2} \times \frac{1}{4}} = \frac{4}{11}$$

26. (d)

$$AM = \frac{\alpha + \beta}{2} = \theta$$

$$\Rightarrow \frac{\cos \alpha + \cos \beta}{\sin \alpha + \sin \beta} = \frac{a}{b}$$

$$\Rightarrow \frac{2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)}{2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)} = \frac{a}{b}$$

$$\Rightarrow \cot\left(\frac{\alpha + \beta}{2}\right) = \frac{a}{b} \Rightarrow \tan \theta = \frac{b}{a}$$

$$\Rightarrow \sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{2b/a}{1 + b^2/a^2} = \frac{2ab}{a^2 + b^2}$$

$$\text{and } \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1 - b^2/a^2}{1 + b^2/a^2} = \frac{a^2 - b^2}{a^2 + b^2}$$

$$\Rightarrow \sin 2\theta + \cos 2\theta = \frac{a^2 - b^2 + 2ab}{a^2 + b^2}$$

27. (c)

$$1 + \tan 1^\circ = 1 + \tan(45^\circ - 44^\circ)$$

$$\tan(45^\circ - 44^\circ) = \frac{\tan 45^\circ - \tan 44^\circ}{1 + \tan 45^\circ \tan 44^\circ} = \frac{1 - \tan 44^\circ}{1 + \tan 44^\circ}$$

$$\tan(45^\circ - 44^\circ) = \frac{\tan 45^\circ - \tan 44^\circ}{1 + \tan 45^\circ \tan 44^\circ} = \frac{1 - \tan 44^\circ}{1 + \tan 44^\circ}$$

$$\text{Hence, } (1 + \tan 1^\circ)(1 + \tan 44^\circ) = 2$$

$$\text{Similarly, } (1 + \tan 2^\circ)(1 + \tan 43^\circ) = 2$$

$$\Rightarrow (1 + \tan 3^\circ)(1 + \tan 42^\circ) = 2 \dots$$

$$(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 43^\circ)$$

$$(1 + \tan 44^\circ)(1 + \tan 45^\circ) = 2^n$$

$$\text{Hence, } 2^{22} \times 2 = 2^n$$

$$2^n = 2^{23} \Rightarrow n = 23$$

28. (c)

$$\text{We know that } \sin(60^\circ - \theta) \sin \theta \sin(60^\circ + \theta) = \frac{\sin 3\theta}{4}$$

$$= \frac{(\sin 48^\circ \sin 12^\circ \sin 72^\circ) \sin 54^\circ}{4 \sin 72^\circ} = \frac{\sin 36^\circ \sin 54^\circ}{4 \sin 72^\circ}$$

$$= \frac{\sin 36^\circ \sin(90^\circ - 36^\circ)}{4 \sin 72^\circ} = \frac{\sin 36^\circ \cos 36^\circ}{4 \sin 72^\circ}$$

$$= \frac{2 \sin 36^\circ \cos 36^\circ}{8 \sin 72^\circ} = \frac{\sin 72^\circ}{8 \sin 72^\circ} = \frac{1}{8} = \frac{1}{2^3} = \left(\frac{1}{2}\right)^3 = \sin^3 30^\circ$$

29. (b)

$$A = 3\hat{i} + 2\hat{j} + \lambda\hat{k}, B = 6\hat{i} + 3\hat{j} + \hat{k}$$

$$C = 5\hat{i} + 7\hat{j} + 3\hat{k} \text{ and } D = 2\hat{i} + 2\hat{j} + 6\hat{k} \text{ are coplanar}$$

$$AB = 3\hat{i} + 5\hat{j} + (1 - \lambda)\hat{k}$$

$$BC = -\hat{i} + 4\hat{j} + 2\hat{k}$$

$$CD = -3\hat{i} - 5\hat{j} + 3\hat{k}$$

$$\text{Now, for coplanar, } \begin{vmatrix} 3 & 5 & (1 - \lambda) \\ -1 & 4 & 2 \\ -3 & -5 & 3 \end{vmatrix}$$

$$\Rightarrow \lambda = 4$$

30. (b)

$$A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i}(2) - \hat{j}(2) + \hat{k}(1)$$

$$A \times B = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$|C - A| = 2\sqrt{2} \Rightarrow |C - A|^2 = 8$$

$$|C|^2 + |A|^2 - 2C \cdot A = 8 \Rightarrow |C|^2 + 9 - 2|C| = 8$$

$$|C|^2 - 2|C| + 1 = 0 \Rightarrow (|C| - 1)^2 = 0$$

$$\Rightarrow |C| = 1$$



Now,

$$\begin{aligned} |(A \times B) \times C| &= |A \times B| |C| \sin \theta \\ &= \sqrt{4+4+1} \times 1 \times \sin 30^\circ \quad [\because \text{Given, } \theta = 30^\circ] \\ &= \frac{3}{2} \end{aligned}$$

31. (a) $AB = 2\hat{i} - 2\hat{j} + \hat{k}$

$AP = 4\hat{i} + \hat{j} - 2\hat{k}$

$$\frac{AB}{|AB|} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}$$

V of any point P (5, -1, -1) on rigid body rotating about axis AB

is given by $V = \left(\left(\frac{AB}{|AB|} \right) \times AP \right) \cdot \omega$

$$\frac{AB}{|AB|} \times AP = \frac{1}{3} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 1 \\ 4 & 1 & -2 \end{vmatrix} = \frac{3\hat{i} + 8\hat{j} + 10\hat{k}}{3}$$

As, $V = \left(\left(\frac{AB}{|AB|} \right) \times AP \right) \cdot \omega$

Hence, $V = \left(\frac{3\hat{i} + 8\hat{j} + 10\hat{k}}{3} \right) \omega = 3\hat{i} + 8\hat{j} + 10\hat{k}$

32. (b) $f(x)$ is a polynomial

such that

$$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

In this case we should know that

$$f(x) = (x)^n \pm 1$$

$$\Rightarrow f(3) = 3^n + 1$$

$$\Rightarrow 28 = 3^n + 1 \Rightarrow 28 = 3^3 + 1$$

$$\Rightarrow n = 3$$

$$f(4) = (4)^3 + 1 = 65$$

33. (c) $A + B = -C$

$$\Rightarrow |A + B| = C$$

On squaring both sides, we get

$$|A|^2 + |B|^2 + 2A \cdot B = |C|^2$$

$$9 + 25 + 2|A||B|\cos \theta = 49$$

$$2|A||B|(\cos \theta) = 49 - 34$$

$$\cos \theta = \frac{15}{2} \times \frac{1}{3} \times \frac{1}{5}$$

$$\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{5\pi}{3}$$

34. Each A_i have 5 elements

Total no. of elements in $A_1 = 5 \times 30 \Rightarrow 150$

$$|S| = \frac{150}{10} = 15 \quad (1)$$

 Each B_i have 3 elements total no. of elements in $B_i = 3n$

$$|S| = 30/9 \quad \dots\dots\dots(ii)$$

$$n = \frac{15 \times 9}{3}$$

$$n = 45$$

35. (c) No. of ways = No. of 1-1 mappings + No. of mappings in which 0 and 1 maps to same element and 2 maps different element + 0 maps to different element and 1 and 2 maps to same element + no. of constant functions

$$= {}^8C_3 + {}^8C_2 + {}^8C_2 + {}^8C_1 = {}^{10}C_3$$

36. (c) $I = \int_0^{\pi/2} \frac{dx}{1+\tan^3 x} = \int_0^{\pi/2} \frac{dx}{1+\frac{\sin^3 x}{\cos^3 x}}$

$$I = \int_0^{\pi/2} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx$$

$$I = \int_0^{\pi/2} \frac{\cos^3(\frac{\pi}{2}-x)}{\cos^3(\frac{\pi}{2}-x) + \sin^3(\frac{\pi}{2}-x)} dx$$

$$I = \int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx$$

Add Eqs. (i) and (ii),

$$2I = \int_0^{\pi/2} \frac{\cos^3 x + \sin^3 x}{\sin^3 x + \cos^3 x} dx$$

$$2I = [x]_0^{\pi/2} \Rightarrow 2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

Or using shortcut formula

$$\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx = \frac{b-a}{2}$$

37. (c) $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n} = \text{Finite non-zero.}$

 Using series of $\cos x$ and e^x ,

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

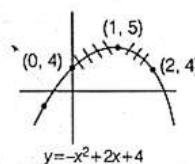
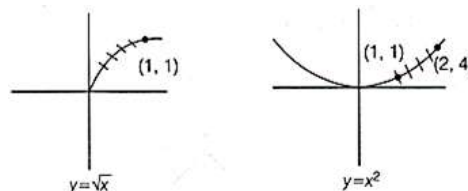
$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\lim_{x \rightarrow 0} \frac{\left[\left(1 - \frac{x^2}{2} + \dots - 1 \right) \right] \left[\left(1 - \frac{x^2}{2} + \dots \right) - \left(1 + \frac{x}{1} + \frac{x^2}{2} + \dots \right) \right]}{x^n}$$

$$= \lim_{x \rightarrow 0} \frac{\left[-\frac{x^2}{2} \right] [-x]}{x^n} = \frac{\left(\frac{x^3}{2} \right)}{x^n} = \left(\frac{1}{2} \right) (x)^{3-n}$$

$$3 - n = 0 \Rightarrow n = 3$$

38. (b)



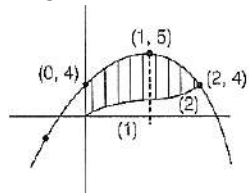
$$A_1 = \int_0^1 \sqrt{x} dx = \left(\frac{2x^{3/2}}{3} \right)_0^1 = \frac{2}{3} = \left(\frac{x^3}{3} \right)_1^1 = \frac{7}{3}$$

$$\begin{aligned} A_3 &= \int_0^2 (-x^2 + 2x + 4) dx \\ &= \left[-\frac{x^3}{3} + x^2 + 4x \right]_0^2 \end{aligned}$$



$$= \frac{-8}{3} + 4 + 8 = \frac{-8}{3} + 12$$

$$= \frac{28}{3}$$



$$A_{\text{final}} = A_3 - (A_1 + A_2)$$

$$= \frac{28}{3} - \left(\frac{2}{3} + \frac{7}{3}\right) = \frac{28}{3} - 3$$

$$= \frac{19}{3} \text{ sq. units}$$

39. (a) Given, $f(x) = 2\sin x + \sin 2x$

$$\Rightarrow f'(x) = 2\cos x + 2\cos 2x$$

For max or min of $f(x)$

$$\Rightarrow f'(x) = 0 \Rightarrow 2\cos^2 x + \cos x - 1 = 0$$

$$\Rightarrow \cos x = \frac{-1 \pm \sqrt{1+8}}{4} = -1, \frac{1}{2}$$

Critical points are $\pi, \frac{\pi}{3}, \frac{5\pi}{3}$

$$f(\pi) = 0, f\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{2}, f\left(\frac{5\pi}{3}\right) = -\frac{3\sqrt{3}}{2}$$

Function has absolute maxima at $x = \frac{\pi}{3}$

Absolute minima at $x = \frac{5\pi}{3}$

40. Let $y = \sec^{-1}\left(\frac{x+1}{x-1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right)$ [$\because \sec^{-1}(a) = \cos^{-1}\left(\frac{1}{a}\right)$]

$$Y = \cos^{-1}\left(\frac{x-1}{x+1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right)$$

$$[\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}]$$

$$Y = \frac{\pi}{2}$$

$$\left[\frac{dy}{dx} = 0\right]$$

REASONING

41. (a)

Step 1 Fill; 9 L container.

Step 2 Pour 4 L water from 9 L container to 4 L container.

Step 3 Empty 4 L container.

Step 4 Again pour 4 L water from 9 L container to 4 L container.

Step 5 Again empty 4 L container.

Step 6 Now pour remaining 1 L water from 9 L container to 4 L container.

Step 7 Fill 9 L container

Step 8 Pour water from 9 L container to 4 L container, so that it is completely full.

Now it is completely full.

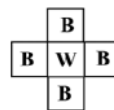
Now, 9 L container contains 6 L water.

42. (d)

The letters are repeat ONE, TWO, THREE, FOUR and FIVE times respectively.

So, the required letter is N.

43. (b)



The required diameter will be diagonal of white square which will give it's middle point as the centre.

So, diameter = $\sqrt{2}$ inch.

44. (b) If x is the amount of fuel consumed while coming back.

$$\text{Then } x + x + \frac{x}{4} = 4\frac{1}{2}$$

$$\Rightarrow \frac{9x}{4} = \frac{9}{2}$$

$$x = 2$$

45. (b)

$$\text{At } x = \frac{9}{11}; \frac{1}{x} = \frac{11}{9} > x:$$

$$\text{At } x = \frac{9}{11}; \frac{x+1}{x} = \frac{20}{9} > x$$

$$\text{At } x = \frac{9}{11}; \frac{x+1}{x-1} = 10 < x$$

Hence, I and II are correct.

46. (d) None of the given answer is correct from the available data set.

$$G = L + 10$$

$$\frac{2A}{3} = \frac{3}{4}\left(B + \frac{A}{B}\right) = \frac{4}{5}\left(G + \frac{1}{4}\left(B + \frac{A}{3}\right)\right)$$

$$= L + \frac{1}{5}\left(G + \frac{1}{4}\left(B + \frac{A}{3}\right)\right)$$

$$\Rightarrow \frac{2A}{3} = \frac{3B}{4} + \frac{A}{4} = \frac{4G}{5} + \frac{B}{5} + \frac{A}{15}$$

$$= L + \frac{G}{5} + \frac{B}{20} + \frac{A}{60}$$

$$\Rightarrow A = 180; B = 100; G = 110; L = 80$$

Which become inconsistent.

47. (c)



Rank from data, top is bottom
Is given.

Hence, R's rank is 5th in the class.

48. (d)

If Bhalu alternated between truth and lie, so will Kachaalu, which will be a contradiction, as only one person alternated between truth and lie.

49. (d)

565xy will be divisible by 80, if $x = 6$ and $y = 0$

$$\Rightarrow x + y = 6$$

50. (b)

7^2 and 3^3 will be factor of

(a. $11^3 \cdot 6^3 \cdot 13^{11}$) If $a = 7^2 \cdot 3 = 147$.



51. (a)

Odd-even = Odd, so $(x - z)^2 y$ will be odd. Hence, (a) cannot be true.

52. (c)

Required distance

$$= 16 + 2 \left[16 \times \frac{1}{2} + 16 \times \frac{1}{2^2} + 16 \times \frac{1}{2^3} + 16 \times \frac{1}{2^4} + \dots \right]$$

$$= 16 + 2 \frac{16 \times \frac{1}{2}}{1 - \frac{1}{2}} = 48 \text{ m}$$

53. (a)

If d is the distance covered, then if 't' hour is the required time to catch the train in time, then

$$\frac{d}{4} = t + \frac{1}{10}$$

$$\text{and } \frac{d}{5} = t = \frac{1}{10}$$

$$\Rightarrow \frac{d}{20} = \frac{1}{5}$$

$$\Rightarrow d = \frac{20}{5} = 4 \text{ km}$$

54. (b) PQSVW is not possible, as S and V cannot be together.

55. (a)

If R is in new office, then as book keeper there is only one option Q. Due to R and U in new office T and S cannot be considered. So, V and W will be sent as new secretaries. So, there is only one option QRUVW.

56. (b)

If R goes, T won't go and if S went further U and V won't go, so we will not get three secretaries.

57. (d)

If S goes to new office, then team will be S, T, W, P, Q only. So, R cannot go and W must go.

58. (a)

	S	T	I	L	L		9	8	5	3	3		
+	W	I	T	H	I	N	⇒	2	5	8	0	5	6
	L	I	M	I	T	S		3	5	6	5	8	9

59. (c)

Total number of hand shakes will be $2(^{12}C_2) = 132$

60. (d)

From given information's, we get

$$Q - U - R - S;$$

$$V > U; P = S + 3$$

⇒ Sequence from the lowest to the highest

I U S Q R P V

61. (a)

If x tricycles are there, then there will be $(10 - x)$ bicycles.

$$3x + 2(10 - x) = 24$$

$$x + 20 = 24$$

$$\Rightarrow x = 4$$

62. (d)

Let the speed of person and that of wind be u and v km/h respectively. d

$$\text{Then } \frac{d}{u - v} = 4$$

$$\text{and } \frac{d}{u + v} = 4$$

$$\Rightarrow 4(u - v) = 3(u + v)$$

$$\Rightarrow u = 7v$$

$$\Rightarrow \frac{d}{u + \frac{u}{7}} = 3$$

$$\Rightarrow \frac{d}{\left(\frac{8u}{7}\right)} = 3$$

$$\Rightarrow \frac{d}{u} = \frac{24}{7}$$

$$\text{Hence, required time} = \frac{24}{7} \text{ h} = 3 \text{ h } 25 \text{ min } 42 \text{ sec}$$

63. (d)

Each 1 in the given series has the digits on it's right exceeding by the digits on the left by

1 1 2; 1 1 2 1 2 2 3;

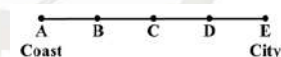
1 1 2 1 2 2 3 1 2 2 3 2 3 3 4

1 1 2 1 2 2 3 1 2 2 3 2 3 3 4 1 2 2 3 2 3 3 4 2

3 3 4 3 4 4 5

So, next three numbers is 4, 3, 4.

64. (c)



The person himself reach the city E from coast A, by travelling through B, C and D each at 30 km each in 4 days.

Then, when returning he will spend his rations at D, where he will be joined by second person with ration of 2 days. At C, third person will be there with ration of 3 persons for one day. Again at B, fourth person will be there, with ration of 4 persons for one day, and hence finally all 4 will reach the coast A.

65. (d)

If y was the total number of eggs and x is the daily sale. Then,

$$\text{Left over egg on day 1} = y - x$$

$$\text{Left over egg on day 2} = 2(y - x) - x = 2y - 3x$$

$$\text{Left over egg on day 3} = 3(2y - 3x) - x = 6y - 10x$$

$$\text{Left over egg on day 4} = 4(6y - 10x) - x = 24y - 41x$$

$$\text{Left over egg or say total eggs on day 5} = 5(24y - 41x) = x$$

$$\Rightarrow 120y - 205 = x$$

$$\Rightarrow 120y = 206x$$



$$\Rightarrow 60y = 103x$$

$$\Rightarrow y = 103 \text{ and } x = 60 \text{ as minimum values.}$$

66. (c) Sujit has told the truth and Lakers will be the winner, so automatically remaining two statements become wrong.

67. (d)

0 9 Case for two digit numbers.

↑

0

↑ - ↑ $9 \times 10 = 0$ cases

9 For three digit numbers.

0

↑↑ — $9 \times 10 = 0$ cases

In 1000 there is 3 0's

Total 0's $9+90+90+3=192$

68. (a)

There is continuation of series.

111 211 211 111 221 312 211 131 122 211 113 213

211 311 312 111 312 211 321 131 112 211 123 113

111 213 211 321 222 111 131 221 133

69. (a)

From the first information oldest daughter will be 9 or 12 yr old.

From the next information her age will be 9.

(AS $9 + 4 = 13$)

70. (a)

Q S C G S C G S C → C G S G S C G S C
 → C G S C G S C S G → C C G G S S C S G
 → C C C S G G S S G → C C C S S G G G S
 → C C C S S G S G G → C C C S S S G G G

Seven moves are shown here out of which one of them is trivial. i.e., six moves are required.

71. (a) 1st Game Mr. Birla defeated Mrs. Birla
 2nd Game Mr. Birla defeated Mrs. Tata
 3rd Game Mrs. Birla defeated Mr. Tata

72. (c)

Second number is the largest. Let it be x , then first number is $x/2$ and third number is $x/3$.

So, their average is

$$\frac{x + \frac{x}{2} + \frac{x}{3}}{3} = 44$$

$$\Rightarrow \frac{11x}{18} = 44$$

$$\Rightarrow x = \frac{44 \times 18}{11} = 72$$

73. (c) 15 large, 7 medium and 14 small ships

$$= 15 \times \frac{(2 \times \frac{7}{4} + 1)}{3} + 14 \text{ small ships.}$$

$$= \frac{105}{4} + \frac{21}{2} + 14 = \frac{203}{4} \text{ small ships.}$$

Now, 12 large, 14 medium and 21 small ships

$$= 12 \times \frac{7}{4} + \frac{14(2 \times \frac{7}{4} + 1)}{3} + 21 \text{ small ships}$$

$$= 21 + 21 + 21 = 63 \text{ small ships}$$

Number of trips by $\frac{203}{4}$ ships = 36

So, number of trips by 63 ships = $\frac{\frac{203}{4} \times 36}{4} = 29$ trips.

74. (a) Q R I P S

According to the given information the order of reading newspaper, so Q passed newspaper to R.

75. (d)

If free luggage is y kg, then Raja has x kg, and Rahim has $2x$ kg extra luggage.

Then, $2y + 3x = 60$

If $3x$ kg extra luggage cost 3600, then 5400 will be cost of

$$\frac{5400}{3600} \times 3x = \frac{9}{2}x$$

$$\Rightarrow y + 3x = \frac{9}{2}x$$

$$\Rightarrow y = \frac{3}{2}x$$

$$\Rightarrow x = 10 \text{ kg}$$

and $y = 15 \text{ kg}$

So, Rahim's luggage is $y + 2x = 35 \text{ kg}$

76. (d)

If x is the number of rows, then if number of children in last row is y , then

$$y + (y + 3) + (y + 6) + \dots + \{y + 3(x - 1)\} = 630$$

$$\Rightarrow xy + \frac{3(x-1)}{2} 630$$

$$x = 3 \Rightarrow 3y + 9 = 630 \text{ (possible, } y = 207)$$

$$x = 4 \Rightarrow 4y + 18 = 630 \text{ (possible, } y = 153)$$

$$x = 5 \Rightarrow 5y + 30 = 630 \text{ (possible, } y = 120)$$

$$x = 6 \Rightarrow 6y + 45 = 630 \text{ (impossible)}$$

Solutions (Q. Nos. 77-80)

City / Player	M	C	K	D	H
P	x	x	x	✓	x
R	x	✓	x	x	x
S	x	x	✓	x	x
U	✓	x	x	x	x
V	x	x	x	x	✓

Game / Player	Cr.	Ch.	Ca.	Ba.	T.T.
P	x	x	x	✓	x
R	x	✓	x	x	x
S	x	x	x	x	✓
U	x	x	✓	x	x



V	✓	×	×	×	✓
---	---	---	---	---	---

77. (a) P
78. (b) Hyderabad
79. (a) Badminton
80. (a) R and Chennai
81. (b) $M \Psi K \$ N \in L$
 $\Rightarrow M$ is mother of K , who is father of N , who is sister of L .
Hence, N is daughter of K .
82. (b)
 $U \Psi R \# S \# T$
 $\Rightarrow R$ is brother of S who is brother of T .
Hence, R is brother of T .
83. (d) $K \# X \Psi Z \# L \$ Y$
 $\Rightarrow X$ is mother of Z , who is brother of father of Y .
Hence, X is real grandmother of Y .
84. (a) $K \Psi L \in M \# N$
 $\Rightarrow K$ is mother of sister of brother of N .
Hence, K is mother of N .
85. (d)
 $M \# N \$ L \# K \$ O$
 $\Rightarrow M$ is brother of father of brother of K who is father of O .
Hence, K is nephew of M .
86. (c)
From the given data following is the possibility.
Anil Lal Babu Sharma Bhatia Gupta
 $\Rightarrow$ Babu has Lal and Sharma as the neighbours.
87. (a)
As the watch gains 10 s in 5 min so it gains 2 min in 1 h. 20 min past 7 O'clock implies 10 h 20 min after 9 am, so actual time is 10 h ahead and 20 min is the gain. Hence, actual time is 7 pm.
88. (c) As by reflexion left hand becomes right hand and vice-versa, so in original clock hour hand is between 8 and
89. (c) Let the ratio be $\lambda : 1$
then, $19\lambda + 9 = (1 + \lambda)$
 $\Rightarrow 4\lambda = 6$
 $\Rightarrow \lambda = \frac{6}{4}$
 $\Rightarrow \lambda : 1 :: 3 : 2$
90. (a)
Each question has 5 options, as it can be answered as A B, C, D or left also.
Now, all 120 questions will have 5 independent options.

Hence, total number of required ways

$$= 5 \times 5 \times 5 \dots \times 5 \text{ (120 times)} = 5^{120}.$$

Computer

91. (b)
 $1 \rightarrow \text{carry}$
 11001100
 $+ 10001111$
 101011011
 $V = 1$, as there is an overflow
 $C = 1$, as carry out is generated by most significant bits
 $Z = 0$, as the result is not zero.
Hence, 1,1,0 is the bits obtained.
92. (b)
A compiler converts a high level program into low level language (Machine language) for execution.
93. (d)
Maximum number
 $= 111111111111 \times 2^{(1111111)^2} = 2^{63}$
94. (c) $4K \times 16 = 2^{12} \times 16$
- ↓ Sign bit

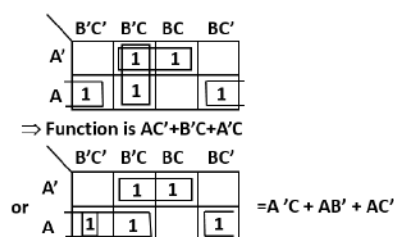
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Mantissa

↓ Sign bit

--	--	--	--	--	--	--	--

Exponent
- $\Rightarrow$ We should have 12 address lines and 16 data lines.
95. (d)
The number of blocks required for the cache increases linealy with the size of the main memory which is the main disadvantage of direct mapping of cache organization.
96. (a) Given, $A[i] = 2i$, where $1 \leq i \leq 10$
 $\Rightarrow A[10] = 2 \times 10 = 20$
Now, $j = A[A[5]] = A[2 \times 5] = A[10] \Rightarrow j = 20$
 $\Rightarrow A[j] = A[20]$
Which is not defined because A takes the values from 1 to 10.
But here, the condition occurs of overflow.
97. (b) BIOS = Basic input Output System.
98. (b) It will be simplified by Karnaugh map as follows.
There are three pairs.




99. (c)

 1's is complement of $1101 = 0010$

 2's complement of $1101 = 0010 + 1 = 0011$

 And 1's complement of $0100 = 1011$

 2's complement of $0100 = 1011 + 1 = 1100$

 Addition both 4 bits 0011

$$\begin{array}{r} 0011 \\ + 1100 \\ \hline 1111 \end{array}$$
 1111

 It's one's complement = 0000

 And 2's complement = $0000 + 1 = 0001$

There is no overflow.

100.(d) $\sqrt{(224)_r} = (13)_r \Rightarrow \sqrt{2r^2 + 2r + 4} = r + 3$

$$\Rightarrow 2r^2 + 2r + 4 = (r + 3)^2 = r^2 + 6r + 9$$

$$\Rightarrow r^2 - 4r - 5 = 0$$

$$\Rightarrow (r - 5)(r + 1) = 0$$

$$\Rightarrow r = -1, 5$$

 As r cannot be -1 , so $r = 5$.

101.(a)

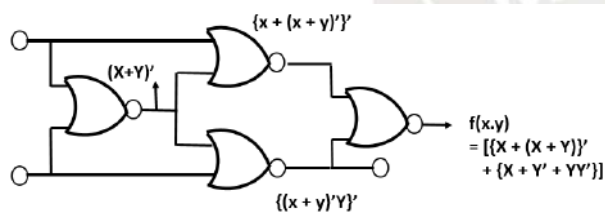
 2's complement of $A = 00000110$

 2's complement of $B = 11110110$
 $11110110 = (246)_{10}$

$$\begin{array}{r} \times 00000110 \\ 00000000 \\ \hline \end{array} = (6)_{10}$$

 11110110
 11110110
 $10111000100 = (1476)_{10}$

 8, bit representation of, 10111000100 is

 $11000100 = (196)_{10}$
102.(b)


$$f(x,y) = [(x + x'.y')' + \{x'.y'+y\}]'$$

(by De Morgan's law)

$$= [x'.(x'.y') + (x'.y')']$$

(By De Morgan's law)

$$= \{x'.(x+y) + (x+y).y'\}'$$

(By De Morgan's law)

$$= \{x'.(x+y)\}' \cdot \{(x+y).y'\}'$$

(By De Morgan's law)

$$= [x + (x+y)'] \cdot [(x+y)' + y]$$

(By De Morgan's law)

$$= [x + x'.y'] \cdot [x'.y' + y]$$

(By De Morgan's law)

$$= (x + x') \cdot (x + y') \cdot (x' + y) \cdot (y' + y)$$

(By De Morgan's law)

$$= 1 \cdot (x + y') \cdot (x' + y) \cdot 1$$

(By complement law)

$$= (x + y') \cdot (x' + y)$$

(By identity law)

$$= x \cdot x' + xy + y'y' + y'y$$

(By Distributive law)

$$= 0 + xy + y'y' + 0$$

(By complement law)

$$= xy + x'y'$$

(By identity and commutative law)

Which represent exclusive NOR.

103.(c)
104.(a)

$$f(x,y,z) = z(x+y) + (\bar{z} + x + y)(\bar{x} + \bar{y}) \text{ (given)}$$

$$= z(x+y) + (z\bar{x}\bar{y}) + (xy) \text{ (By De Morgan's Law)}$$

$$= xy + yz + zx + z\bar{x}\bar{y} \text{ (By Distributive law)}$$

$$= yz + zx + (xy + z\bar{x}\bar{y})$$

$$= yz + zx + (xy + z)(\bar{x} + \bar{y}) \text{ (By Distributive law)}$$

$$= yz + zx + (xy + z) \cdot 1 \text{ (By complement law)}$$

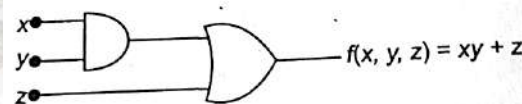
$$= yz + zx + xy + z$$

$$= zx + xy + z \cdot 1$$

(By Absorption law)

$$= xy + z(x + 1) = xy + z \cdot 1 = xy + z$$

$$\Rightarrow f(x,y,z) = xy + z$$



It requires two gates.

105.(a)

ENGLISH

106. (b)	107. (c)	108. (d)
109. (d)	110. (a)	111. (c)
112. (c)	113. (a)	114. (d)
115. (d)	116. (c)	117. (b)
118. (a)	119. (b)	120. (c)



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